

ERICK DAVID CEDILLO PÉREZ

SOFTWARE ENGINEERING | EMBEDDED SYSTEMS | CYBERSECURITY

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PROFESSIONAL SUMMARY

Software Engineer with 3 years of practical experience in embedded systems, full-stack development, and cybersecurity focused on governance. Over the last 12 months, served as Tech Lead for an institutional web platform serving 50+ active users, conducted banking infrastructure audits as a Cybersecurity Consultant under ISO 27001 standards, and delivered technical instruction to 45 students. Prioritizing clean architecture, real-world performance optimization, and engineering-grade documentation.

WORK EXPERIENCE

Systems Development Tech Lead | ENP 6 – UNAM

October 2025 – Present

- **Full-Stack Architecture:** Architected and deployed a multi-purpose native PHP/JS system (user management, laboratories, scheduling, inventory, and loans) actively serving 50+ users in production.
- **Data Resilience:** Designed and implemented a custom fault-tolerance mechanism based on JSON Journaling.
- **Technical Documentation:** Authored a comprehensive 50-page engineering documentation manual covering user guides, system maintenance, and architectural chronicles.

Cybersecurity Consultant | Law Firm (Banking Client)

May 2025 – February 2026

- **Vulnerability Assessment:** Executed infrastructure audits using Nmap, documenting 3–4 critical vulnerabilities across mail servers and local network environments.
- **Governance & Compliance:** Designed and implemented 15+ information security governance policies aligned with the ISO 27001 standard.
- **Executive Reporting:** Translated complex technical findings into actionable operational reports for non-technical banking stakeholders.

Front-End Instructor | ENP 6 – UNAM

May 2025 – June 2025

- **Technical Training:** Designed and taught an intensive 4-week JavaScript curriculum (ES6+, DOM Manipulation, Asynchronous Programming).
- **Retention & Mentorship:** Achieved a 96% student retention rate (43/45 students) by conducting targeted 1-on-1 mentorship sessions.

ENGINEERING PROJECTS

WALL-E (C++ / ESP32)

- Autonomous waste-collector robot featuring mapping navigation, object detection, and obstacle avoidance.
- Efficient memory management using Circular Buffers and hardware protection via dual voltage regulators.

Automated Bridge (Python / Raspberry Pi 5)

- Scale-model system powered by Python control logic, touchscreen HMI, and 3D mechanical modeling in Autodesk Inventor.
- Technical documentation structured at thesis project level.

C RPG Engine (C / Allegro)

- Engine-less 2D RPG developed in native C. Built frame animations, hitboxes, knockback physics, pathfinding AI, dialogue trees, and boss battles.

EDUCATION

B.S. in Electrical and Electronics Engineering | School of Engineering, UNAM

In Progress

Associate Degree in Software Development | ENP 6, UNAM

Graduated 2025

TECHNICAL SKILLS

Languages:	C, C++, Python, PHP, JavaScript (ES6+), Bash, Java, Visual Basic
Hardware / Embedded:	ESP32, Raspberry Pi 5, Arduino, PWM, I2C, Power Regulation
Cybersecurity:	Nmap, IT Governance, ISO 27001, Infrastructure Auditing
Infra & Tools:	Git, SQL, Linux SysAdmin, WireGuard, Tailscale, Docker, Figma
Languages (Spoken):	Spanish (Native), English (B2)